# Power Series Solutions to Nonlinear Ordinary Differential Equations
# MATH 689
# Homework 6

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April 14, 2026

**1.** For each of the following series, apply Pringsheim's theorem (and corollaries) to the pattern indicated in the terms provided to determine locations of the singularities of $f$ responsible for divergence. Assume that the eventual pattern does not change as further terms are included.

**(a)**

$$\begin{aligned} f(x) = 1 + 2x + \frac{5x^2}{2} + \frac{8x^3}{3} + \frac{65x^4}{24} + \frac{163x^5}{60} + \frac{1957x^6}{720} + \frac{685x^7}{252} + \frac{109601x^8}{40320} + \frac{98641x^9}{36288} \\ + \frac{9864101x^{10}}{3628800} + \frac{13563139x^{11}}{4989600} + \frac{260412269x^{12}}{95800320} + \frac{8463398743x^{13}}{3113510400} + \frac{47395032961x^{14}}{17435658240} \\ + \frac{888656868019x^{15}}{326918592000} + \frac{56874039553217x^{16}}{20922789888000} + \frac{7437374403113x^{17}}{2736057139200} + \frac{17403456103284421x^{18}}{6402373705728000} \\ + \frac{82666416490601x^{19}}{30411275102208} + \cdots, \quad |x| < 1. \end{aligned} \tag{1}$$

*Solution.* By Pringsheim's Theorem where the coefficients $a_n$ of a representable function $f(z)$ via the power series $\sum a_n z^n$ are positive for some $n > p$, or $n \to \infty$. Then a singularity of $f(z)$ lies at the point $z = R$ lies on $\mathbb{R}^+$.

Given that we have positive coefficient with a radius $R = 1$. We say there is a singularity at $z_s = R = 1$

**(b)**

$$\begin{aligned} f(x) = \frac{1}{2} + \frac{x}{4} + \frac{x^2}{8} + \frac{x^3}{48} + \frac{x^4}{96} - \frac{x^5}{960} + \frac{7x^6}{5760} - \frac{41x^7}{80640} + \frac{43x^8}{161280} - \frac{383x^9}{2903040} \\ + \frac{1919x^{10}}{29030400} - \frac{21101x^{11}}{638668800} + \frac{63307x^{12}}{3832012800} - \frac{117569x^{13}}{14233190400} + \frac{5760889x^{14}}{1394852659200} \\ - \frac{86413319x^{15}}{41845579776000} + \frac{86413321x^{16}}{83691159552000} - \frac{1469026453x^{17}}{2845499424768000} + \frac{13221238081x^{18}}{51218989645824000} - \cdots, \quad |x| < 2. \end{aligned} \tag{2}$$

*Solution.* By Corollary 2.3.3 for Coefficients That Eventually Alternate. We can rewrite (2) with $R = 2$

$$\sum_{0 \le n \le 8} R^n a_n \left(\frac{z}{R}\right)^n + \sum_{9 \le n \le \infty} R^n a_n \left(\frac{z}{R}\right)^n = \sum_{0 \le n \le 8} R^n a_n \left(\frac{z}{R}\right)^n + \left(\frac{z}{R}\right)^9 \left(\frac{1}{1 - \left(\frac{z}{R}\right)}\right) \tag{3}$$

From this it is easy to see that the first 8 term is a continuous polynomial and the singularities comes from the infinite series at $z_s = -R = -2$ which lies within the radius.

**(c)**

$$f(x) = \frac{1}{16} + \frac{x^4}{256} + \frac{x^8}{4096} + \frac{x^{12}}{65536} + \frac{x^{16}}{1048576} + \frac{x^{20}}{16777216} + \frac{x^{24}}{268435456} + \cdots, \quad |x| < 2. \tag{4}$$

*Solution.* By Corollary 2.3.4 for Series That (Eventually) Are the Same Sign and Skip Terms. We can represent the series with $\lambda = 4$ as

$$f(z) = \frac{1}{16} \sum_{0 \le n \le \infty} \left(\frac{z^\lambda}{16}\right)^n, \quad z_{s,} = R\cos\left(\frac{2m\pi}{\lambda}\right) + iR\sin\left(\frac{2m\pi}{\lambda}\right), \quad m \in [\lambda - 1] \tag{5}$$

Where $[\lambda - 1]$ denotes the integers from 0 to $\lambda - 1$ inclusive. This yields the singularities $z_{s0} = 2, z_{s1} = 2i, z_{s2} = -2, z_{s3} = -2i$ in this respective order for $m = 0, 1, 2, 3$.

**2**. For the power series solution to the Flierl-Petviashvili problem that you found in Problem 2 of HW3, apply Pringsheim's theorem (and corollaries) to the $a_n$ coefficients to estimate locations of the singularities of $f$ responsible for divergence. For this, you will need the radius of convergence $R$ estimated in HW3.

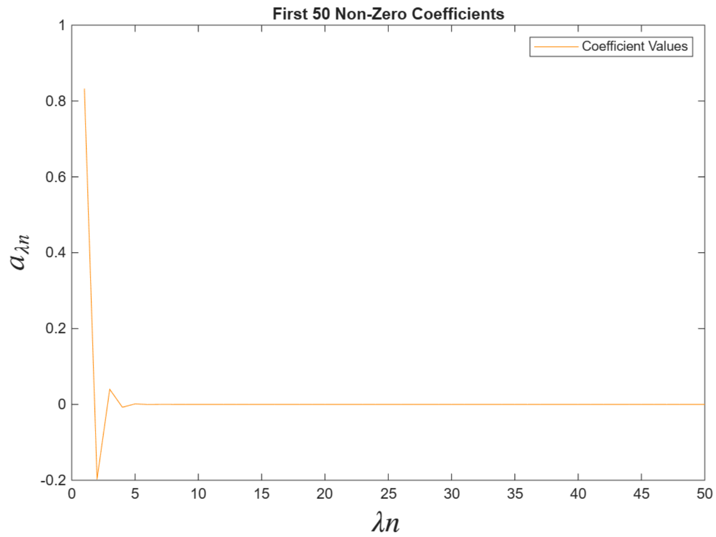



Figure 1: First 50 Non-Zero Coefficients

*Solution.* We notice that the values of the coefficients are skipping and alternating signs. For example the first 10 values of the coefficients are

$$\{a_n\}_{0\leq n\leq 10} = \{-2.392, 0, 0.832416, 0, -0.196866384, 0, 0.039940522, 0, -0.007482569, 0, 0.001335647\} \tag{6}$$

It follows that we use Corollary 2.3.5 Series That (Eventually) Skip Terms and Alternate in Sign. Using the value of $R = 2.58409$ and $\lambda = 2$ yields the singularities of the form

$$z_{sm} = R\exp\left(\frac{i(2m+1)\pi}{\lambda}\right) = R\cos\left(\frac{(2m+1)\pi}{\lambda}\right) + iR\sin\left(\frac{(2m+1)\pi}{\lambda}\right), \quad m \in [\lambda - 1] \tag{7}$$

Thus $z_{s0} = iR = 2.58409i, \quad z_{s1} = -iR = -2.58409i$

**3.** The shape of a meniscus with contact angle $\theta$ at a flat wall in an infinite liquid pool is described in Batchelor (1967) [1] by the ODE

$$\frac{dh}{dx} = -2\left[\left(h^2-2\right)^{-2} - \frac{1}{4}\right]^{1/2}, \qquad h(0) = \sqrt{2(1-\sin\theta)}, \qquad 0 \leq x < \infty \tag{8}$$

where $h(x)$ is the meniscus height at a distance $x$ away from the wall.

**(a)** Modify one of your existing codes to solve the above problem numerically over the domain $x \in [0,1]$. Use $\theta = \pi/4$ for this assignment.

*Solution.* From $\theta = \pi/4 \Rightarrow h(0) = \sqrt{2-\sqrt{2}}$ A plot is shown in part **(d)**.

**(b)** The power series solution was developed in Naghshineh et al. (2023) [2]. Show that it is given by

$$\begin{aligned}
h &= \sum_{n=0}^{\infty} a_n x^n, \qquad a_{n+1} = \frac{-2d_n}{n+1}, \qquad a_0 = \sqrt{2(1-\sin\theta)}, \\
d_{n>0} &= \frac{4}{n(4c_0-1)}\sum_{j=1}^{n}\left(\frac{3}{2}j - n\right)c_j d_{n-j}, \qquad d_0 = \sqrt{c_0 - 1/4}, \\
c_{n>0} &= \frac{1}{n(b_0-2)}\sum_{j=1}^{n}(-j-n)\,b_j c_{n-j}, \qquad c_0 = (b_0-2)^{-2}, \\
b_n &= \sum_{j=0}^{n} a_j a_{n-j}.
\end{aligned} \tag{9}$$

*Solution.*
To derive the solution given above, let $h = \sum_{0\le n\le\infty} a_n x^n$

$$\begin{aligned}
\frac{dh}{dx} = \sum_{0\le n\le\infty}(n+1)a_{n+1}x^n &= -2\left\{\left[\sum_{0\le n\le\infty}\left(\sum_{1\le j\le n} a_{n-j}a_n\right)x^n - 2\right]^{-2} - \frac{1}{4}\right\}^{\frac{1}{2}} \\
&= -2\left[\left(\sum_{0\le n\le\infty} b_n x^n - 2\right)^{-2} - \frac{1}{4}\right]^{1/2} \\
&= -2\left[\sum_{0\le n\le\infty}\left(\frac{1}{n(b_0-2)}\sum_{1\le j\le n}(-j-n)\,b_j c_{n-j}\right)x^n - \frac{1}{4}\right]^{1/2} \\
&= -2\left[\left(\sum_{0\le n\le\infty} c_{n>0}x^n\right) - \frac{1}{4}\right]^{1/2}, \qquad c_0 = (b_0-2)^{-2} \\
&= -2\left[\left(\sum_{0\le n\le\infty} c_{n>0}x^n - 1/4\right)\right]^{1/2} \\
&= -2\sum_{0\le n\le\infty}\left[\frac{4}{n(4c_0-1)}\sum_{j=1}^{n}\left(\frac{3}{2}j - n\right)c_j d_{n-j}\right]x^n, \qquad d_0 = \sqrt{c_0 - 1/4} \\
&= -2\sum_{0\le n\le\infty} d_{n>0}x^n
\end{aligned} \tag{10}$$

Thus we can write the recurrence as

$$a_{n+1} = \frac{-2d_{n>0}}{n+1}, \qquad a_0 = h(0) = \sqrt{2(1-\sin(\theta))} \tag{11}$$

**(c)** Incorporate the power series solution into your code.
*Solution / Code.* See Attached
**(d)** Plot successive truncations of the series on the same graph as the numerical solution. Confirm that they agree in some region near $x = 0$. Use $N = 20$ and skip every 4 terms.

*Solution / Plot.*

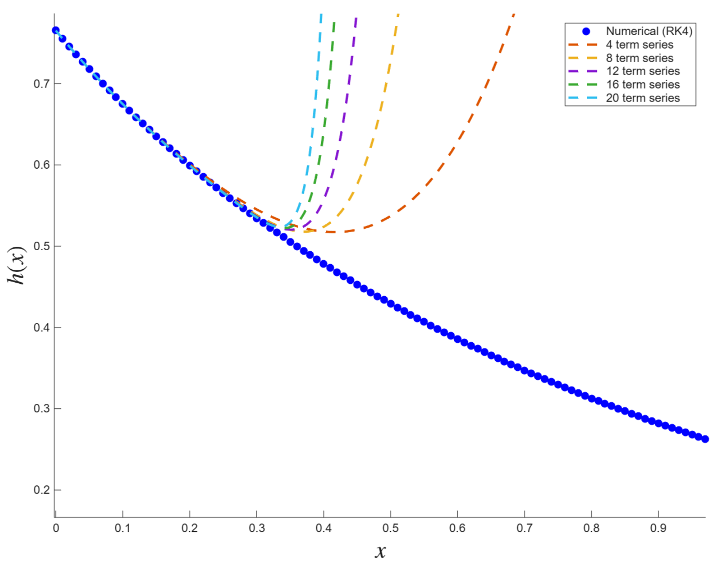



Figure 2: Power Series Solution and RK4 for $h(x)$

For $x \approx 0$ we can see that the numerical solution closely matches the power series solution.

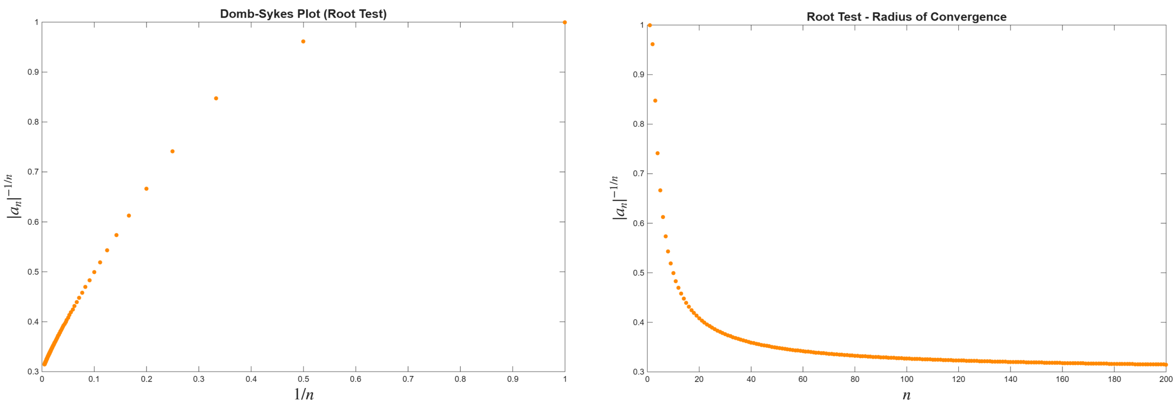
**(e)** Examine the numerical values of the coefficients in MATLAB and use Pringsheim's theorem (and corollaries) to show that the power series diverges due to a singularity on the negative real $x$-axis.

*Solution.* Looking at the first 11 terms of the sequence we can again see that the coefficients are alternating in signs.

$$\{a_n\}_{0\leq n\leq 10} \approx \{0.7653, -1, 0.72159, -0.7440, 0.49603, -0.1155, -0.0656, -0.3461, 1.75354, -4.3090\} \tag{12}$$

By Corollary 2.3.5, we there exist a singularity of $h(z)$ at the point $z_s = -R$.
**(f)** Use the numerical implementation of the root and/or ratio test to estimate the radius of convergence $R$. Combine this knowledge with (e) to locate the singularity of $h(x)$ closest to $x = 0$.

*Solution / Plot.*




(a) Root Domb-Sykes Plot

(b) Root Test Plot

Figure 3: Radius Estimates: $R \approx 0.302239670316083$

**(g)** We may resum the series via Eulerization as follows:

$$h = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} A_n \left( \frac{x}{x+R} \right)^n \tag{13}$$

where

$$A_{n>0} = \sum_{m=1}^{n} \binom{n-1}{m-1} a_m R^m, \qquad A_0 = a_0. \tag{14}$$

Incorporate the Eulerized series into your code using the snippet provided.

*Solution / Code.* See Attached

**(h)** Plot truncations of the Eulerized series on the same plot as the original power series and the numerical solution. Confirm that the Eulerized series analytically continues the original power series.

*Solution / Plot.*

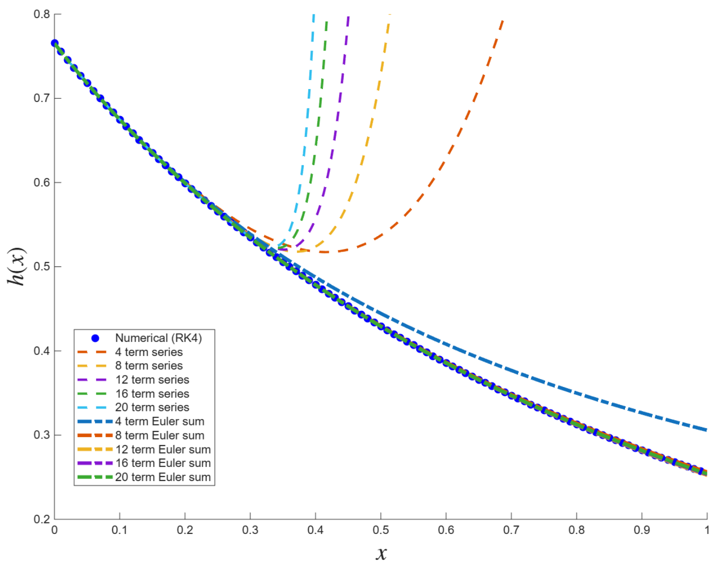



Figure 4: Analytic Continuation

**4. (MATH689 Problem): Eulerization of the Flierl-Petviashvili (FP) Problem** Since the closest singularity of the FP problem is not on the negative real $x$-axis, a straight Eulerization in $x$ will not work.

**(a)** Looking at the pattern deduced in Problem 2, explain why the variable substitution $u = x^2$ will lead to a series in $u$,

$$f = \sum_{n=0}^{\infty} b_n u^n, \qquad b_n = a_{2n}, \tag{15}$$

that does have a singularity on the negative real $u$-axis, such that an Eulerization

$$f = \sum_{n=0}^{\infty} b_n u^n = \sum_{n=0}^{\infty} B_n \left(\frac{u}{u+R^2}\right)^n \tag{16}$$

will work to analytically continue the series.

*Solution.* From Problem 2 we can see that $\lambda = 2$, meaning that the recurrence is only nonzero every two terms, hence we can write $a_{2n} = b_n, n \geq 0$. Also note that if we use the substitution $u = x^2$ we can obtain

$$f = \sum_{0 \leq n \leq \infty} a_{2n}(x^{2n}) = \sum_{0 \leq n \leq \infty} b_n u^n \tag{17}$$

Also the pattern for $b_n$ becomes

$$\{b_n\}_{0 \leq n \leq 5} = \{-2.392, 0.832416, -0.196866384, 0.039940522, -0.007482569, 0.001335647\} \tag{18}$$

then by Corollary 2.3.5 we know that a singularity of $f$ lies on the point $z_s = -R = -2.58409$

**(b)** Incorporate the Eulerization of the transformed series into your existing FP code and confirm that the Eulerized series analytically continues the original power series.

*Solution / Plot.*

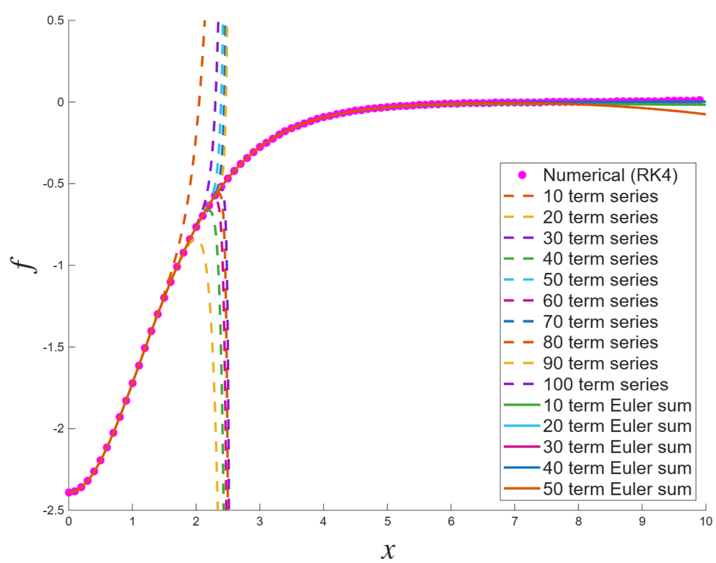



Figure 5: Eulerized Solution

**(c)** How does this Eulerized gauge resummation compare with the ordinate resummation that you did for this problem in HW5?

*Solution.* The Ordinate Resummation uses a dependent variable substitution. Explicitly, we used $F(\cdot) = (\cdot)^{-1}$ on another series, so that

$$f = \sum_{0 \leq n \leq \infty} a_n x^n = \frac{1}{\sum_{0 \leq n \leq \infty} A_n x^n} \tag{19}$$

On the other hand, the Eulerized Gauge resummation makes a substitution on the independent variable.

# References

[1] George Keith Batchelor. *An introduction to fluid dynamics.* Cambridge university press, 2000.

[2] Nastaran Naghshineh, W Cade Reinberger, Nathaniel S Barlow, Mohamed A Samaha, and Steven J Weinstein. On the use of asymptotically motivated gauge functions to obtain convergent series solutions to nonlinear odes. *IMA Journal of Applied Mathematics*, 88(1):43–66, 2023.